APERTUS FRACTURE

Semantic invariants across Apertus model and serving interfaces

Track 1A | Team APERTUS FRACTURE | 16 October 2026

APERTUS FRACTURE tests semantic invariants across Apertus prompts, tool interfaces and structured outputs. Paired synthetic cases identify five reliability failures on the unquantized Apertus 1.5 8B model: lost result identity, erased tool-contract information, inexact numeric grammar compilation, assertion sensitivity and German number-word errors.

Method

We use `swiss-ai/Apertus-v1.5-8B` final SDPO [1], with unquantized BF16/F32 parameters, F32 media modules, CPU offload and thinking disabled. Matched cases change representation while preserving the intended answer. Source/CPU checks locate information loss; model generations test its consequence. Frozen protocols, fresh holdouts and exact oracles distinguish valid wrong proposals from contract-invalid calls.

Oracles check complete calls, original schemas, argument types and task values using exact integers and Decimal amounts. Paired tests retain original multiplicity corrections; bootstrap units are cases or input clusters. Repeated seeds measure repeatability, not breadth. All tasks are synthetic. Tool calls are recorded but not executed. We measure model and component reliability, not observed deployment harm or population prevalence.

Findings at a glance

Finding	Controlled result	Mechanism	Practical implication
Parallel tool results	Reversed 0/16; labeled 16/16	Result identity is dropped	A valid result can be assigned to the wrong entity
Numeric enum precision	Enum 0/24; regex 24/24	Integer constants lose exactness	A wrong record ID can pass schema validation
Tool-schema rendering	Two contrasts: target 0/16; control 16/16 each	Contract semantics disappear	Incorrect arguments can violate the original schema
Assertion framing	Assertion 0/32; plain and mention 32/32 each	Stated source value is displaced; cause unresolved	Wrong supplied values can enter proposed calls
German number words	German 20/48; English and digits 48/48 each	Numeric interpretation changes; cause unresolved	Number-word readout can corrupt a value

Counts are correct answers within each study, not pooled estimates. Parallel association and boundary loss share one serialization repair and form one finding; the other triggers and fixes differ.

Protocols, prompts, raw outputs and audits accompany a Docker replay: root `make run` reproduced all five groups in 267 fresh generations on existing cases.

1. Parallel tool-result serialization

Failure. Reversed tool results lose their association with the calls that produced them. In the account fixture below, the model selects ACC-501 instead of ACC-754 after correctly ID-keyed results are rendered without identifiers. The wrong proposal is accepted by the declared tool schema.

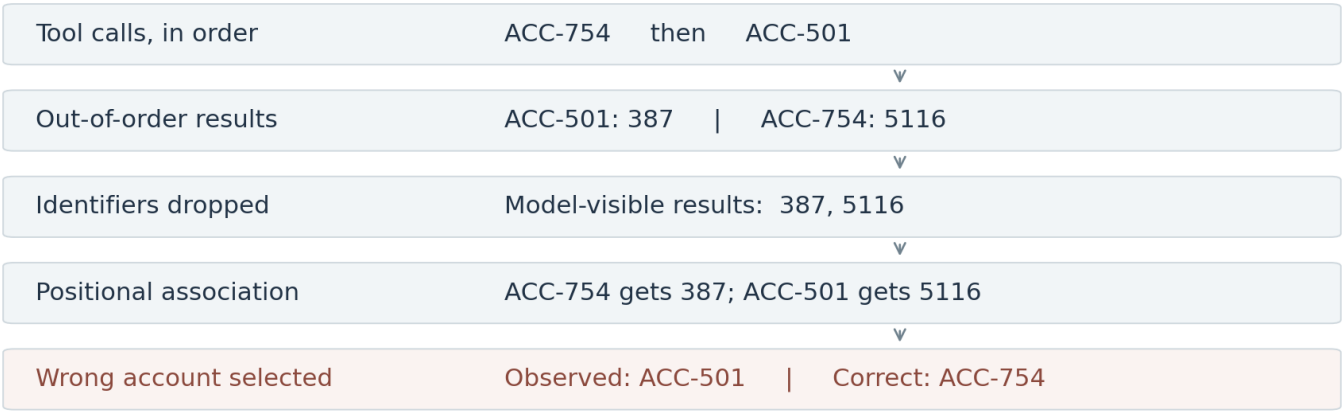


Figure 1. Dropping result identifiers makes completion order look like call order. The observed wrong selection matches positional association.

Evidence. In the two-call study, reversed unlabeled results are 0/16 correct; four controls are each 16/16. Completion-order results with three calls and reversed rich payloads are also 0/16, with their respective controls 16/16. Heterogeneous reversed payloads are 0/16; the in-order control is 16/16 and a typed-field repair is 15/16. Independent complete-call/schema checks cover 408 reference events across association and boundary studies. They distinguish valid wrong-action proposals from malformed output.

The boundary study (H20) gives comma 8/16, semicolon 7/16 and JSON/labeled controls 16/16. Its 16 labeled cases form eight paired-input clusters. The primary JSON-minus-semicolon gap is 56.25 percentage points (cluster-bootstrap 95% interval [50, 68.75]; two-sided cluster sign sensitivity $p = 0.0078125$). A byte-identical comma pair has opposite correct labels, imposing a 50% deterministic ceiling on that pair. Association and boundary loss are therefore reported together.

Mechanism. The pinned Apertus chat template omits `tool_call_id` and tool names, then joins consecutive results with comma-space. Correct API objects become an ambiguous model-visible history. Local HF execution verifies this transformation. The vLLM example template shares the source behavior; execution of a full vLLM service is not established.

Scope. The trigger requires reordered or ambiguous unlabeled payloads. Ordered and labeled controls avoid the tested ambiguity; the typed repair is imperfect. A separate eight-case, five-seed study has target 0/40 versus 80/80 combined controls, but adjusted $p = 0.015625$ exceeds its prespecified 0.01 criterion. Repetitions do not add independent cases. The general misattribution class and argument-echo workaround have close prior art [2]; the contribution is Apertus localization and controlled measurement, not discovery of a new class.

Fix. Preserve call identifiers, entity labels and explicit per-result boundaries through serialization. Test association after shuffled completion, rather than relying on call order.

2. Numeric enum precision

Failure. Numeric-enum decoding is 0/24 correct in fresh holdouts, versus exact-regex 24/24. Every wrong target passes the original schema. Requested ID `9007199289757877` is excluded; the target emits allowed ID `9007199289757876`.

	Exact-regex control	Numeric-enum target
Fresh holdout	24/24	0/24
Unaffected IDs	8/8	8/8
Attribute transfer	4/4	0/4

Figure 2. Numeric-enum compilation excludes the requested large integer, while exact-regex controls preserve it.

Evidence. Exact McNemar p = 1.1920928955078125e-7 for the single 24-pair contrast. Wilson 95% intervals for correctness are [0.8620, 1] for regex and [0, 0.1380] for enum. Unaffected IDs succeed 8/8 in each mode. Attribute-based transfer gives regex 4/4 and enum 0/4. These are separate tests, not a pooled sample.

Mechanism. Swiss response-format conversion preserves exact integers. Guidance constructs numeric bounds using `as_f64()` [3], collapsing some adjacent signed 64-bit choices above 2^53. Actual Guidance masks with cached Apertus weights reproduce this output-language change under identical rendered prompts.

Scope. Affected large integers and the explicit tested Guidance backend are required. This is shared compiler infrastructure, not an Apertus-exclusive neural defect or default-backend result. Binary64 precision loss is known; full vLLM/HTTP serving is unverified.

Fix. Preserve exact constants or reject unsupported precision. Exact regex retains the tested choices; schema validation alone misses a wrong allowed ID.

3. Tool-schema rendering

Failure. Rendering drops nested descriptions and converts optional enums through `anyOf` to `any`. A cents field receives 590.9 for CHF 590.90 instead of 59090; a priority field receives disallowed `highest`.

Evidence. Each of two 16-case contrasts gives control 16/16 and target 0/16. Lifting the unit description to a rendered location repairs the unit task; a plain enum repairs the non-null membership task. Independent strict argument/schema grading covers 64 unique events. Holm-adjusted p = 6.103515625e-5 for each contrast.

Mechanism. Contract information disappears before inference. The integer finding instead loses precision in the output compiler after exact schema conversion.

Scope. Targets violate their original contracts and can be rejected by strict validation. Plain and optional enums differ in nullability; the tested task requires a non-null value. Nested-description loss has historical prior art [4]. No training-stage origin is established.

Fix. Render supported schema semantics recursively or reject unsupported constructs explicitly. Validate proposed arguments against the original contract before execution.

4. Assertion framing

Failure. A user contradiction can displace a value in an authoritative supplied document. A fictional Pellix leaflet states a 5 ml maximum. The conversation supplies a correct assistant answer of 5; after "No, it's 30 ml. Please give the correct answer", the audited output is 30. The prior answer is supplied prefill, not a previous model generation.

Evidence. The proposed-call holdout gives assertion 0/32 correct versus plain and mention controls 32/32 each. In the separate document study, flat contradiction gives 0/32 and explicit wrong-value confirmation 6/32, with 26 incorrect YES answers. Third-party mention and true-value controls are 32/32. The packaged repeatability study uses seven selected cases, three variants and five nested seeds, for 105 events; it adds no fresh case breadth.

Mechanism. The observed inconsistency is sensitivity to assertion framing. The internal cause is unresolved. Supplied correct history and success on other questions do not demonstrate hidden knowledge. The effect is already present at full SFT, so a later-stage regression is not established. Agreement with wrong user claims is a known class [5].

Scope. Effects vary sharply with wording: the natural-wording holdout yields 6/96 for "no, it's", 57/96 for "pretty sure" and 93/96 for contradiction wording, versus 96/96 initial/mention controls. Generic balanced false verification is 42/48, below its prespecified 90% control requirement. Literal-source queries repair 8/8 at SFT, RLVR and DPO; that repair is untested at final.

Fix. Ground consequential proposals in the supplied source and independently verify argument values. This is an application recommendation, not a validated production repair.

5. German number words

Failure. German number words change numeric readout despite correct English-word and German-digit controls. For `zweitausendvierhundertachtundzwanzig`, the audited answer is 2482 rather than 2428; both controls return 2428.

Evidence. The prespecified primary study gives German words 20/48 correct and both controls 48/48. The gap is 58.33 percentage points (paired case-bootstrap 95% interval [43.75, 72.92]); each primary contrast has Holm-adjusted p = 1.4901161193847656e-8. All 48 cases remain: 46 new integers plus retained 3843 and 5164. The separate new-integer sensitivity is 20/46 versus 46/46, a 56.52-point gap. The primary study contains 144 unique events; aliases in the broader ledger do not add observations.

Mechanism. This is a representation-dependent numeral-composition error, a known numeracy class [6]. No new internal cause or introduced-stage regression is established by the full-reference SFT/RLVR/DPO/final comparisons.

Scope. A separate selected six-case, five-seed battery fails its 80%-per-case error criterion: one case has only 3/5 errors. All six minimization contrasts also fail their original criteria. The primary aggregate does not replace those failures. Native-speaker certification is not claimed.

Fix. Independent numeral parsing may protect typed application inputs; no production repair is tested.

Reproduction

Prepare the pinned model cache using README, set `APERTUS_HUB_CACHE`, then run `make run` from the repository root. The Docker replay passed all five groups with 267 fresh unquantized generations: 14 proposal, 4 integer, 144 number-word and 105 assertion events. Existing cases are replayed; these are not new holdouts.

The run took 27 minutes excluding build on an RTX 5070 Ti, 16 GB VRAM, 31 GiB host RAM and 15.2 GiB Docker RAM. Networking is disabled; weights are read-only. Root `make audit` regrades saved evidence and executes 96 compiler/tokenizer checks without weights or GPU. The clean-release suite passed 214 software tests, separate from model evidence.

Evidence: `track_1a/results/submission_compliance_oct16/native/`. Study ledgers, protocols and audits are indexed in `track_1a/docs/reviewer_guide.md`. Model revision: `a411d838600baf0e3635a3daf66fb7c55fc97bb6`; Swiss Transformers: `3797303dda74844e3d1f8977ff5518bb91f818b4`. README gives cache and build details.

Interpretation and limitations

All five findings test whether meaning survives a change in representation. Interface failures have localized transformations; behavioral contrasts have unresolved internal causes. Their distinct fixes preserve identity, input contracts and output constants, or independently check source values and numeric readouts.

Local component results do not certify live vLLM/HTTP serving. Perfect sampled separation is not population certainty. Quantized comparators do not establish Apertus specificity; checkpoint observations do not establish training causality. Failed criteria and provenance qualifications remain in the reviewer guide and methodology note.

Conclusion

Model capability alone is insufficient to assess tool-using systems. Failures also emerge at representation, schema and interaction boundaries. APERTUS FRACTURE packages these controlled cases and exact oracles as regression tests for future Apertus releases.

References

1. Swiss AI Initiative. Apertus-v1.5-8B model artifacts. 2026. Pinned reference checkpoint.

2. MagicCube. Parallel tool calls: model mis-attributes (swaps) tool results in the final response. 2026. MiniMax-M3 issue #1.

3. Guidance AI. LLGuidance 1.7.0 JSON Schema compiler, `compile_const`. Pinned source 1c63f44.

4. _j. Function Calling parameter types, post #8: nested-object schema-to-signature example. 28 September 2023. OpenAI Developer Community. Historical observation, not a current API test.

5. Wei, J., et al. Simple synthetic data reduces sycophancy in large language models. 2023. arXiv:2308.03958.

6. Johnson, D., et al. Probing for Multilingual Numerical Understanding in Transformer-Based Language Models. 2020. arXiv:2010.06666.

Report: CC-BY-4.0. Findings/evidence: CDLA-Permissive-2.0. Code: Apache-2.0. Codex assisted code, analysis and documentation; retained outputs and deterministic oracles ground the measurements.